Improved Elastic Weight Consolidation as an Optimization Constraint for Continual Learning

Davide Wiest
Department of Computer Science, TU Darmstadt; Independent Researcher
Darmstadt, Germany
2026

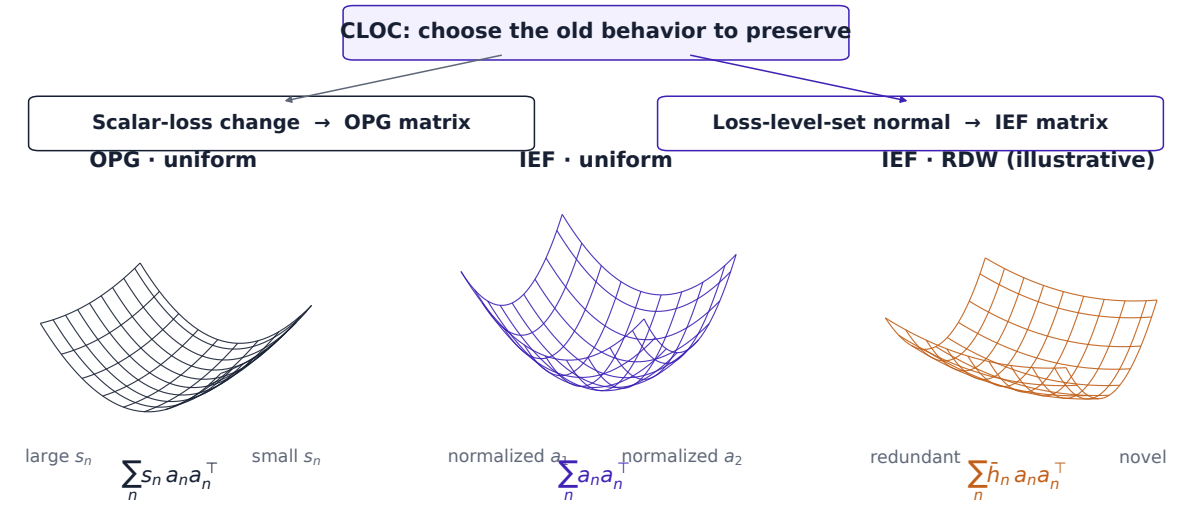


Figure 1: Each surface shows the continual-learning penalty over normalized directions, which locally approximate loss changes for two representative samples. Observed-label OPG/EWC induces anisotropy through gradient-scale weighting; uniform IEWC removes this scaling away from the damping regime, while the separate RDW variant weights samples by streaming residual diversity. [1]

Abstract

Elastic Weight Consolidation (EWC) is widely used in continual learning, but its importance matrix is implemented in several inequivalent ways. In particular, the observed-label outer product often called the empirical Fisher is neither the model Fisher nor a general curvature matrix. We derive this object instead from old-solution preservation. Starting from continual learning as an optimization constraint (CLOC), we introduce a population, low-order relaxation for constructing scalable local penalties. Preserving each old sample's scalar loss yields an outer product of loss gradients (OPG); for negative log likelihood, this algebraically coincides with the observed-label empirical Fisher used by EWC. Preserving outputs modulo motion tangent to the old loss level set instead yields the improved-empirical-Fisher (iEF) normalization and leads to IEWC. This derivation isolates the mechanism: OPG scales each local sensitivity direction by the squared output-loss-gradient magnitude, whereas undamped iEF removes that scale. We show exactly how positive damping restores attenuation near zero gradient, and introduce task-relative damping that preserves invariance to global loss units. In a controlled contamination study, corrupted samples dominate the OPG trace and rotate its leading directions, while

iEF strongly reduces their trace share and remains aligned with the clean-contribution eigenspace. We evaluate the resulting diagonal penalties with locked test sets, paired task orders, matched global scale, and separate uniform and residual-diversity weighting, so that evidence for normalization is not attributed to a weighting heuristic.

Code: https://github.com/Axym-Labs/iewc
Project page: https://axym.org/work/iewc
Correspondence: dw@axym.org

1 Continual Learning as an Optimization Constraint

Continual learning asks a model to learn from sequentially available distributions without unrestricted access to earlier data [2–5]. After training on distribution A and obtaining $\theta_A^\star$, minimizing only the later loss $L_B(\theta)$ leaves old behavior unconstrained. Methods such as EWC, SI, MAS, and LwF address this problem with different preservation objects: parameter importance, learning-path importance, output sensitivity, or old predictions [6–9]. This variety raises a prior question: *what quantity should a local continual-learning penalty preserve?*

We pose that question directly as constrained optimization. Let $q : \Theta \to \mathcal{H}$ be a quantity whose old value or acceptable set represents preservation, and let $\mathcal{C} \subset \mathcal{H}$ encode the corresponding constraint. Learning distribution B becomes

$$\min_{\theta \in \Theta} L_B(\theta) \qquad \text{subject to} \qquad q(\theta) \in \mathcal{C}.$$

We call this perspective *Continual Learning as an Optimization Constraint* (CLOC). CLOC is deliberately a formulation rather than a single algorithm: different choices of q, $\mathcal{C}$, relaxation, and approximation recover different local preservation penalties.

The paper develops one path through this design space. First, a population and low-order relaxation turns samplewise preservation objects into a stored quadratic penalty. Second, preserving a scalar old-sample loss yields an outer-product loss-gradient (OPG) matrix; for negative log likelihood, this is algebraically the observed-label empirical Fisher used in common EWC implementations. Third, preserving output motion only in the normal quotient of the old loss level set yields the iEF normalization and a geometric interpretation of IEWC. We make damping, sample weighting, output metric, and diagonal approximation explicit instead of treating them as consequences of the normalization. Finally, controlled mechanism tests and validation-selected, locked evaluations ask which parts of this construction survive finite damping and improve continual learning in the studied domains.

This framing preserves the original intuition—continual learning as old-solution preservation—while separating exact local algebra from global curvature claims and empirical questions. The next section constructs the practical relaxation and states its feasibility and locality boundaries.

2 Related Work and Scope

EWC stores a quadratic approximation around earlier parameters and weights parameter motion by a Fisher-derived importance estimate [6]. Subsequent work has clarified how multiple task penalties should be accumulated and placed EWC within broader Bayesian and regularization-based views of continual learning [10–12]. A practical ambiguity remains: implementations called "Fisher" may use the model-label expectation, sampled model labels, observed labels, or minibatch aggregates. These estimators are not interchangeable in theory or practice [13, 14]. Our scalar-loss derivation concerns the observed-label OPG object; sampled model-Fisher EWC is evaluated separately rather than treated as another name for the same matrix.

Other regularization methods protect different objects. SI estimates parameter importance from the optimization trajectory, MAS uses sensitivity of the learned output function, and LwF distills old predictions on later inputs [7–9]. Functional regularization can instead preserve selected predictive functions or memorable points [15, 16]. CLOC is complementary to this taxonomy: it begins by declaring a preservation object and constraint set, then asks which local stored penalty follows. The present paper studies scalar-loss preservation and loss-level-set-normal output preservation; it does not claim that these are the only useful choices or that replay, distillation, and parameter isolation share the same resource assumptions.

iEF was introduced as a normalized observed-gradient approximation for natural-gradient optimization [17]. We recover its Euclidean algebra from a different question: the metric distance of an old output displacement from the tangent to its old loss level set. This provides a preservation interpretation but does not convert iEF into exact global curvature. Our additions relative to simply substituting iEF into EWC are the CLOC derivation, the matched output-metric generalization, the finite-damping and interpolation boundary, and the separation of normalization from aggregation weights. The RDW ablation borrows a streaming residual-diversity intuition from GSS [1]; it is an explicitly order-dependent heuristic, not part of the level-set result.

3 Population and Local Approximations

For a closed convex set $\mathcal{C}$, the Moreau envelope of its indicator is a scaled squared distance [18]. This motivates the soft CLOC objective

$$\min_\theta J(\theta) = \min_\theta L_B(\theta) + \frac{\lambda}{2}\operatorname{dist}(q(\theta), \mathcal{C})^2 = \min_\theta L_B(\theta) + \frac{\lambda}{2}\inf_{u\in\mathcal{C}} \|q(\theta) - u\|^2.$$

GEM and A-GEM are similar in spirit in that they formulate preservation through constraints on old examples [19, 20]. CLOC turns the old-task constraint into a penalty built after task-A training. In a population setting, we use sample-based analogues ℓ_B and q and take the empirical average. Define the unscaled preservation loss as $p(a, \theta) = \operatorname{dist}(q(a, \theta), \mathcal{C})^2$.

$$\min_\theta \frac{1}{N_B}\sum_{i=1}^{N_B} \ell_B(b_i, \theta) + \frac{\lambda}{2N_A}\sum_{i=1}^{N_A} p(a_i, \theta)$$

The convex Moreau–Yosida identity motivates this penalty form but is not a global theorem for every set used below. Neural-network loss level sets are generally nonconvex, their projections need not be unique, and squared distance need not be globally smooth. Our operative construction is local: we replace each exact set by its tangent model at the old output and derive the resulting quadratic penalty. We do not require a globally convex old-loss level set.

The samplewise population form is also stronger than preserving only expected old risk: it asks every retained sample surrogate to remain near its own constraint. Those equalities can conflict with learning the new distribution or with each other after approximation. The penalty parameter λ is therefore a soft feasibility–plasticity tradeoff, not a guarantee that all hard constraints remain exactly satisfiable.

The two-distribution presentation is only for notational simplicity; additional old distributions can be handled by adding analogous constraints. We refer to this population version as **population CLOC (P-CLOC)**. One efficiency consideration is the penalty p: Ideally, we let the constraint contribute to each gradient step with its gradient. However, recomputing the full old-distribution penalty p and its associated factors many times during optimization on distribution B is usually prohibitively expensive, since it depends on the dataset D_A associated with A. Additionally, reusing dataset D_A later on does not respect the continual learning setting. For this reason, replace the exact penalty by a local

k-th order Taylor-polynomial approximation around the old solution $\theta_A^\star$. This turns the computation of p into a repeated evaluation of precomputed derivative tensors and bounds memory requirements to $O(d^k)$ for a d-dimensional parameterization. Since memory scales exponentially with k, the standard scalable surrogate is a low-order Taylor approximation. We refer to this local approximation as **low-order CLOC (L-CLOC)**, and to its population version as **PL-CLOC**. It remains to specify the preserved quantity q and constraint set $\mathcal{C}$. Notice that the gradient of p is dependent on only one sample at once, which allows us to analyze p and its gradients on a per-sample basis.

4 Old-Task Loss Preservation

Since ℓ_A is a practical surrogate for performance on distribution A, a natural choice is to use the loss itself to define preservation. We say that f_θ locally preserves an old sample if it preserves that sample's old loss value. This gives

$$q(a, \theta) = \ell_A(f_\theta(x)) - \ell_A(f_{\theta_A^\star}(x)).$$

Since preservation means $q(a, \theta) = 0$, take $\mathcal{C} = \{0\}$. The corresponding unscaled proximal loss is

$$p(a, \theta) = \mathrm{dist}(q(a, \theta), \mathcal{C})^2 = q(a, \theta)^2.$$

To obtain a low-order Taylor approximation, use the perturbation around the old solution $\delta = \theta - \theta_A^\star$ in the first-order expansion of q:

$$q(a, \theta_A^\star + \delta) \approx g^\top \delta,$$

where

$$g = \nabla_\theta \ell_A(f_{\theta_A^\star}(a)).$$

Equivalently, writing

$$z_A = f_{\theta_A^\star}(a) \qquad r_i = \nabla_z \ell_A(z)\big|_{z=f_{\theta_A^\star}(a)}, \qquad J_i = \nabla_\theta f_{\theta_A^\star}(a),$$

the chain rule gives

$$g_i = J_i^\top r_i.$$

Substituting this expansion into the preservation penalty gives

$$\frac{\lambda}{2N_A} \sum_{i=1}^{N_A} q(a_i, \theta)^2 \;\approx\; \frac{\lambda}{2N_A} \sum_{i=1}^{N_A} (g_i^\top \delta)^2.$$

Collecting the gradient outer products yields

$$\frac{\lambda}{2N_A} \sum_{i=1}^{N_A} (g_i^\top \delta)^2 = \frac{\lambda}{2} \delta^\top \left(\frac{1}{N_A} \sum_{i=1}^{N_A} g_i g_i^\top \right) \delta = \frac{\lambda}{2} \delta^\top M_1 \delta.$$

Therefore the resulting matrix is

$$M_1 = \frac{1}{N_A} \sum_{i=1}^{N_A} g_i g_i^\top = \frac{1}{N_A} \sum_{i=1}^{N_A} J_i^\top r_i r_i^\top J_i.$$

This matrix encodes the local parameter directions in which the chosen scalar preservation quantity changes. We call it the *outer-product loss-gradient* (OPG) matrix. The construction is exact for the squared first-order surrogate displayed above, not for the original nonlinear equality constraint. It becomes the observed-label empirical-Fisher algebra only for likelihood losses, as specified next.

5 Outer-Product Loss-Gradient Regularization and EWC

The matrix obtained in the previous section is

$$M_{\mathrm{OPG}} = \frac{1}{N_A} \sum_{i=1}^{N_A} g_i g_i^\top .$$

It is an outer product of gradients of the selected scalar preservation object. This identity holds for any differentiable loss, without a probabilistic model or a curvature interpretation. We therefore call the general object an *outer-product-of-loss-gradients* (OPG) matrix.

For a negative-log-likelihood loss,

$$\ell_i(\theta) = -\log p_\theta(y_i \mid x_i),$$

we have $g_i = -\nabla_\theta \log p_\theta(y_i \mid x_i)$. The same OPG algebra then coincides with the observed-label score-gradient matrix commonly called the empirical Fisher,

$$F_{\mathrm{emp}} = \frac{1}{N_A} \sum_{i=1}^{N_A} \nabla_\theta \log p_\theta(y_i \mid x_i) \nabla_\theta \log p_\theta(y_i \mid x_i)^\top .$$

Thus OPG-EWC and empirical-Fisher EWC denote the same implementation in an explicit likelihood setting. The CLOC derivation justifies this matrix as the quadratic form of a squared first-order loss-preservation surrogate. It does not turn the matrix into the model Fisher or establish that it approximates the Hessian.

The model Fisher is instead

$$F_{\mathrm{model}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}_A} \mathbb{E}_{y \sim p_\theta(\cdot \mid x)} \left[\nabla_\theta \log p_\theta(y \mid x) \nabla_\theta \log p_\theta(y \mid x)^\top \right].$$

It averages over labels sampled from the model, not only the observed label. Under standard regularity conditions this Fisher is connected to expected negative-log-likelihood curvature and natural-gradient geometry [21–23]. The observed-label empirical Fisher is a different object and can be a poor curvature approximation [13]; Fisher-estimator choice can also materially affect EWC behavior and regularization scale [14].

This distinction separates two questions that are often conflated. First, why might the observed-label OPG be a useful continual-learning importance matrix? The local CLOC construction answers: it penalizes first-order changes of old sample losses. Second, is that matrix a faithful estimate of probabilistic Fisher curvature? Not in general. Our experiments therefore include a one-label Monte Carlo model-Fisher baseline, obtained by sampling from the frozen predictive distribution at each supervised output site—one class per image and one token per supervised TRACE answer position—alongside OPG-EWC.

The word "exact" below always refers to the algebra of the squared first-order local surrogate. The original hard per-sample constraint, its population counterpart, and the nonlinear loss away from $\theta_A^\star$ remain approximated.

6 Level-Set Preservation in Output Space

Our first mechanism is derived from preservation of the loss value. An extension of the same preservation constraint is to impose it through the old-task loss level set in output space. More concretely, we preserve the output per-sample up to motions that stay on this set. This output-space perspective connects to distillation and functional regularization approaches to continual learning [9, 15, 16, 24], which let the model match old predictions or predictive functions. This gives a geometric formulation of the same underlying constraint: instead of penalizing raw change in the scalar loss coordinate,

we penalize displacement away from the local constraint set in output space. Let $a \in D_A$ be an old sample, and write $z_A = f_{\theta_A^\star}(x)$. The exact old-loss preservation set in output coordinates is

$$\mathcal{C} = \{z : \ell_A(z) = \ell_A(z_A)\}.$$

Thus the preserved quantity is the model output itself:

$$q(a, \theta) = f_\theta(x).$$

The hard constraint is

$$q(a, \theta) \in \mathcal{C},$$

and the corresponding unscaled proximal loss is

$$p(a, \theta) = \operatorname{dist}_G\left(q(a, \theta), \mathcal{C}\right)^2,$$

where $G \succ 0$ is the chosen local output metric. Positive definiteness makes the primal metric and its inverse on covectors unambiguous; Section 11 treats the restricted pseudoinverse convention needed by a singular ordered-output example.

This exact penalty is generally not tractable. To obtain a practical form, use the standard first-order local model of the constraint set near the old output z_A, namely its tangent plane. Let

$$r = \nabla_z \ell_A(z)\big|_{z=z_A}.$$

We first assume $r \neq 0$, so the loss level set has a regular tangent hyperplane at z_A. For a small output displacement $\Delta z = f_\theta(x) - f_{\theta_A^\star}(x)$, the first-order old-loss change is approximately $r^\top \Delta z$. Hence the local constraint set is

$$T = \{\Delta z : r^\top \Delta z = 0\}.$$

Replacing the exact level set by this tangent hyperplane gives the local proximal loss

$$p(a, \theta) \approx \operatorname{dist}_G(\Delta z, T)^2.$$

For the metric $|v|_G^2 = v^\top G v$, the squared distance to the tangent hyperplane follows from projecting Δz onto the gradient:

$$\operatorname{dist}_G(\Delta z, T)^2 = \frac{(r^\top \Delta z)^2}{r^\top G^{-1} r}.$$

To verify the matched metric pair, write the normal residual as $e = \Delta z - t$ with $t \in T$. It solves

$$\min_e e^\top G e \quad \text{subject to} \quad r^\top e = r^\top \Delta z.$$

The Lagrange equations give

$$e^\star = \frac{r^\top \Delta z}{r^\top G^{-1} r} G^{-1} r,$$

whose squared G-norm is the displayed quotient. Thus G measures primal output displacement while G^{-1} acts on the loss covector; these roles cannot be chosen independently. Linearizing the model output around $\theta_A^\star$, with

$$\delta = \theta - \theta_A^\star, \qquad J = \nabla_\theta f_{\theta_A^\star}(x),$$

we have

$$\Delta z \approx J\delta.$$

Substituting this into the local regularization term gives

$$\frac{\lambda}{2} p(a, \theta_A^\star + \delta) \approx \frac{\lambda}{2} \frac{(r^\top J \delta)^2}{r^\top G^{-1} r}.$$

Equivalently,

$$\frac{\lambda}{2} p(a, \theta_A^\star + \delta) \approx \frac{\lambda}{2} \delta^\top \left(J^\top \frac{r r^\top}{r^\top G^{-1} r} J \right) \delta.$$

Taking the empirical average over old-distribution samples $a_1, \ldots, a_N$ yields an importance matrix of the form

$$M_{2,A} = \frac{1}{N} \sum_{i=1}^{N} J_i^\top \frac{r_i r_i^\top}{r_i^\top G_i^{-1} r_i} J_i.$$

This is a local construction: it penalizes motion normal to the tangent model, not distance to a globally convex neural-network level set. When $r_i = 0$, the tangent normal is undefined and the undamped quotient is $0/0$; our finite-damping implementation assigns that sample zero contribution. Section 10 makes the resulting near-interpolation boundary explicit.

7 Level-Set Normalization and the Improved Empirical Fisher

In the Euclidean-output case $G_i = I$, the matrix from the previous section is the improved empirical Fisher (iEF) introduced by Wu et al. [17]:

$$M_{2,I} = F_{\text{IEF}}.$$

Hence iEF is the quadratic form of a first-order local constraint that protects old behavior geometrically: instead of preserving a raw scalar-loss coordinate, it penalizes only output motion normal to the old loss level set. The original iEF paper derives the same normalization for approximate natural-gradient descent. In that setting, the observed-label OPG reweights sample contributions by squared output-gradient magnitude; iEF removes this scale factor and retains the normalized output-loss direction. Here the normalization arises from metric projection onto the normal quotient of the old-loss tangent space.

This is an identity for the undamped local summand when $r_i \neq 0$. Practical damping changes its behavior near $r_i = 0$; Section 10 states that boundary explicitly.

8 Relation to Gauss-Newton

Consider the scalar least-squares setting in Euclidean geometry. Suppose the model output is scalar, such that

$$\ell_n(z) = \frac{1}{2}(z - y_n)^2.$$

Then

$$r_n = z_n^A - y_n$$

is scalar, and for $r_n \neq 0$ we have

$$\frac{r_n^2}{r_n^2} = 1.$$

Therefore the *undamped* Euclidean IEF contribution reduces to

$$J_n^\top J_n,$$

and averaging gives

$$M_{\text{IEF}} = \frac{1}{N} \sum_{n=1}^{N} J_n^\top J_n.$$

This is exactly the Gauss–Newton or output-Jacobian form for squared-error regression [22, 25, 26], the standard practical preconditioning matrix for second-order optimization. This matches our geometric intuition: loss preservation is governed by how parameters move the output, which is encoded by the Jacobian J, not by the current output-gradient magnitude. This equivalence was made explicit in [17]. With positive damping, the contribution is instead multiplied by $r_n^2/(r_n^2+\tau)$ and tends to zero at exact interpolation; the exact Gauss–Newton identity therefore requires $\tau = 0$ and $r_n \neq 0$.

9 Normalization and Sample Weighting

To compare both matrices in a common notation, let $G_n \succ 0$ denote the chosen output metric for sample n. Although M_{OPG} itself does not depend on G_n, its summands can be rewritten relative to that metric. In a negative-log-likelihood setting, $M_{\mathrm{OPG}} = F_{\mathrm{emp}}$.

Define

$$\hat{r}_n^G = \frac{r_n}{\sqrt{r_n^\top G_n^{-1} r_n}}, \qquad a_n^G = J_n^\top \hat{r}_n^G, \qquad w_n^G = r_n^\top G_n^{-1} r_n.$$

Then, wherever $r_n \neq 0$,

$$M_{\mathrm{IEF},G} = \frac{1}{N}\sum_{n=1}^{N} a_n^G (a_n^G)^\top, \qquad M_{\mathrm{OPG}} = \frac{1}{N}\sum_{n=1}^{N} w_n^G\, a_n^G (a_n^G)^\top.$$

If we interpret both matrices as empirical aggregates of local old-task sensitivity directions $a_n^G(a_n^G)^\top$, they use the same directions but differ in how strongly each sample contributes. Undamped IEF keeps uniform empirical weighting of these normalized directions, while OPG multiplies each contribution by the metric-dependent squared magnitude

$$w_n^G = r_n^\top G_n^{-1} r_n.$$

Thus normalization changes sample aggregation: OPG shifts old-task weighting toward samples with large output-loss gradients and away from confidently fitted ones, whereas undamped IEF removes that scale. With positive damping, IEF also multiplies each direction by $s_n/(s_n+\tau)$, as Section 10 makes explicit.

This decomposition identifies a mechanism to test; it does not show that either weighting is preferable for every task. The next section treats probabilistic classification, where the scale difference has a concrete well-fitted limit.

10 Gradient-Scale Effects in Classification

The output-gradient weighting in observed-label OPG has an important classification symptom. For cross-entropy with softmax probabilities p_n and one-hot label y_n, the output-loss gradient with respect to logits is

$$r_n = p_n - y_n.$$

For a confidently correct prediction with class c and small ε,

$$p_{n,c} = 1-\varepsilon, \qquad \|r_n\|_2^2 \in \Theta(\varepsilon^2).$$

Thus the OPG contribution becomes small as the model becomes more confident and correct. This is coherent with its derivation: the scalar first-order loss change is also small. For continual-learning preservation, however, small current loss need not mean that losing the sample later is harmless. EWC-DR identifies this classification symptom and proposes a logit-reversal repair [27].

In the present framework, that observation is one instance of the weighting effect identified in Section 8: observed-label OPG importance depends on output-gradient scale. Undamped IEF removes this particular factor. Positive damping retains the removal only above its transition scale, and empirical benefit remains a question for the controlled comparisons below.

11 Behavior Near Well-Fitted Solutions

The output-loss-gradient scale in OPG is especially consequential because importance is estimated *after* training on A, when many samples may already be well fit. The original EWC Laplace argument likewise expands at an old-task optimum, although its model-Fisher/Hessian motivation is distinct from the observed-label OPG object analyzed here [6, 13]. Modern expressive models can fit training data up to interpolation [28], so the near-zero regime must be treated explicitly rather than hidden by notation.

To make the comparison precise, consider the per-sample summands

$$S_n^{\mathrm{OPG}} = J_n^\top r_n r_n^\top J_n, \qquad S_n^{\mathrm{IEF},G} = J_n^\top \frac{r_n r_n^\top}{r_n^\top G_n^{-1} r_n} J_n,$$

where $G_n \succ 0$ is the chosen output metric. We assume that G_n is uniformly positive definite and bounded, and that J_n is bounded under the operator norm. More precisely, assume there exists a constant $C_J < \infty$ such that

$$\|J_n^\top v\|_2 \le C_J \|v\|_2$$

for the relevant directions, and constants $0 < m_G \le M_G < \infty$ such that G_n is uniformly bounded between $m_G I$ and $M_G I$.

Using the same G_n-normalized direction as above,

$$\hat{r}_n^G = \frac{r_n}{\sqrt{r_n^\top G_n^{-1} r_n}}, \qquad a_n^G = J_n^\top \hat{r}_n^G, \qquad w_n^G = r_n^\top G_n^{-1} r_n, \qquad b_n^G = \|J_n^\top \hat{r}_n^G\|_2^2,$$

This gives

$$S_n^{\mathrm{IEF},G} = a_n^G (a_n^G)^\top, \qquad S_n^{\mathrm{OPG}} = w_n^G \, a_n^G (a_n^G)^\top.$$

For the spectral norm,

$$\|S_n^{\mathrm{OPG}}\|_2 = w_n^G b_n^G, \qquad \|S_n^{\mathrm{IEF},G}\|_2 = b_n^G.$$

Since

$$\|\hat{r}_n^G\|_2^2 = \frac{\|r_n\|_2^2}{r_n^\top G_n^{-1} r_n},$$

the metric bounds and bounded-Jacobian assumption give $b_n^G \in O(1)$, so

$$\|S_n^{\mathrm{OPG}}\|_2 \in O(w_n^G), \qquad \|S_n^{\mathrm{IEF},G}\|_2 \in O(1).$$

The sharper lower-bound statement requires a nondegeneracy condition in the normalized output-gradient direction. If there exists $c_J > 0$ such that

$$\|J_n^\top \hat{r}_n^G\|_2 \ge c_J$$

for the relevant samples, then

$$\|S_n^{\mathrm{OPG}}\|_2 \in \Theta(w_n^G), \qquad \|S_n^{\mathrm{IEF},G}\|_2 \in \Theta(1).$$

We now interpret this for practical situations. Near the end of training on distribution A, we expect many old samples to have small gradients. The norm analysis above then shows that OPG contains

the additional multiplicative factor $w_n^G = r_n^\top G_n^{-1} r_n$, whereas undamped IEF removes that factor. Remaining low-magnitude summands of the IEF come from the model sensitivity term b_n^G, not from the scale of the output-loss gradient itself. In the classification setting with $G_n = I$, a confidently correct prediction with $p_{n,c} = 1 - \varepsilon$ yields

$$\|r_n\|_2^2 = \|p_n - y_n\|_2^2 \in \Theta(\varepsilon^2).$$

Therefore

$$\|S_n^{\mathrm{OPG}}\|_2 = w_n^I b_n^I \in \Theta(\varepsilon^2)\, b_n^I, \qquad \|S_n^{\mathrm{IEF},I}\|_2 = b_n^I.$$

Under the same nondegeneracy condition on $J_n^\top \hat{r}_n^I$, this gives $\|S_n^{\mathrm{OPG}}\|_2 \in \Theta(\varepsilon^2)$ and $\|S_n^{\mathrm{IEF},I}\|_2 \in \Theta(1)$.

The singular case $r_n = 0$ remains and must be handled explicitly. As done in [17], one practical solution is to add a damping factor τ to the quotient, yielding

$$M_{\mathrm{IEF},G} = \frac{1}{N} \sum_{n=1}^{N} \frac{g_n g_n^\top}{r_n^\top G_n^{-1} r_n + \tau}, \qquad g_n = J_n^\top r_n.$$

For $s_n = r_n^\top G_n^{-1} r_n > 0$, each damped summand admits the exact factorization

$$S_{n,\tau}^{\mathrm{IEF},G} = \frac{g_n g_n^\top}{s_n + \tau} = \underbrace{\frac{s_n}{s_n + \tau}}_{\rho_\tau(s_n)} S_{n,0}^{\mathrm{IEF},G}.$$

Thus finite damping introduces a transition rather than a uniform scale removal. When $s_n \gg \tau$, $\rho_\tau(s_n) \approx 1$ and the undamped direction-normalization analysis applies. When $s_n \ll \tau$, $\rho_\tau(s_n) \approx s_n/\tau$, so small output-loss gradients are again suppressed. Since $g_n = J_n^\top r_n$, fixed positive damping gives $S_{n,\tau}^{\mathrm{IEF},G} \to 0$ as $r_n \to 0$ under a bounded Jacobian. At exact interpolation, $r_n = g_n = 0$; we define the sample contribution as zero. The undamped expression is instead $0/0$ and has no canonical value without specifying a directional limit.

Fixed numerical τ also has the units of s_n. Under a global rescaling of one task loss by c, the numerator and s_n scale by c^2, but a fixed τ does not. We therefore use task-relative damping in the confirmatory experiments:

$$\tau_A = \kappa \ \mathrm{median}\{s_i : s_i > 0\}, \qquad \kappa = 10^{-2}.$$

Both s_n and τ_A then scale by c^2, giving

$$\frac{c^2 g_n g_n^\top}{c^2 s_n + c^2 \tau_A} = \frac{g_n g_n^\top}{s_n + \tau_A}.$$

Task-relative damping therefore restores invariance to global changes of loss units while retaining explicit attenuation near zero gradient. It does not imply invariance to arbitrary sample-dependent monotone transformations under a shared positive damping value. If all recorded s_i are zero, the task importance is defined as the zero matrix.

12 Output-Space Geometry

The scalar CLOC construction and its OPG matrix require no output-space metric. The level-set construction does: for a symmetric positive-definite primal metric G_n, output displacements have inner product $\langle u, v \rangle_{G_n} = u^\top G_n v$, the metric-normal direction is $G_n^{-1} r_n$, and the normal-distance denominator is $r_n^\top G_n^{-1} r_n$. Thus a geometry choice must supply a matched pair: G_n acts on primal output

displacement and G_n^{-1} on the loss covector. An arbitrary denominator does not constitute a new geometry.

The canonical choice $G_n = I$ recovers iEF [17]. Other choices can represent a local quadratic notion of output stability, but the assumptions matter. A positive-semidefinite ambient metric is sufficient only after specifying a tangent space or quotient, a Moore–Penrose pseudoinverse, and the relevant range conditions. Diagonalizing the resulting parameter pullback can further discard cross-parameter geometry.

As one explicit non-Euclidean example, consider probability vectors on an ordered grid with m bins. Let

$$P = I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top$$

project onto the probability-simplex tangent, let $C \in \mathbb{R}^{(m-1)\times m}$ compute cumulative sums through the penultimate bin, and let Δx be the grid spacing. Define

$$G_{\mathrm{CDF}} = \Delta x\, PC^\top CP.$$

For a zero-sum probability perturbation Δz,

$$\Delta z^\top G_{\mathrm{CDF}} \Delta z = \Delta x\, \|C\Delta z\|_2^2,$$

the squared L^2 distance between the corresponding one-dimensional cumulative distribution functions. The matched dual form is

$$(Pr)^\top G_{\mathrm{CDF}}^{+}(Pr).$$

Our implementation tests both identities numerically. This is an ordered-CDF or Cramér metric, not sliced Wasserstein. We retain the earlier diffusion experiment only as exploratory evidence that alternative output representations can be inserted into IEWC; it is not used to claim that the diagonal experiment validates Wasserstein geometry.

13 Notation and Approximation Boundaries

Table 1 collects the objects that are otherwise easy to conflate.

Symbol or name	Meaning
$g_i = J_i^\top r_i$	Parameter gradient of the selected old-sample scalar loss.
M_{OPG}	Mean $g_i g_i^\top$ from scalar-loss preservation; only the likelihood-score case is the observed-label empirical-Fisher algebra.
F_{model}	Fisher information averaging score outer products over labels sampled from the frozen predictive distribution.
$s_i = r_i^\top G_i^{-1} r_i$	Squared dual norm of the output-loss gradient under the matched output metric.
M_{IEF}	Level-set-normalized mean $g_i g_i^\top/(s_i + \tau_A)$; the undamped Euclidean case is iEF.
$\rho_\tau(s)$	Damping attenuation $s/(s+\tau)$ relative to the undamped summand.
κ	Dimensionless relative-damping coefficient in $\tau_A = \kappa\,\mathrm{median}_{s_i>0}\, s_i$.
$\bar{h}_i$	Optional mean-one aggregation weight; uniform weighting and RDW are evaluated separately.
Γ	Rule assigning the local primal output metric G_i.

Table 1: Notation and distinctions used throughout the paper.

Our derivations concern full positive-semidefinite matrices, whereas the training experiments store only their diagonals. Taking $\mathrm{diag}(M)$ preserves each coordinate-axis quadratic penalty, nonnegativity of the entries, and the trace. It generally does not preserve off-axis quadratic forms, rank,

eigenvectors, parameterization behavior, or the full normal-space geometry. Consequently, the full-matrix contamination analysis in Appendix B tests the proposed normalization mechanism, while benchmark training separately tests whether its diagonal projection is useful.

Appendix A gives the exact scope of loss-rescaling invariance and Appendix B analyzes and directly tests the contamination mechanism.

14 Improved Elastic Weight Consolidation

The preceding derivation leaves four implementation choices: global penalty strength, damping, aggregation weights, and output geometry. For old-task samples $a_1, \ldots, a_N$, IEWC uses

$$M_{\mathrm{IEWC}} = \frac{1}{N} \sum_{n=1}^{N} \bar{h}_n \frac{g_n g_n^\top}{r_n^\top G_n^{-1} r_n + \tau_A}, \qquad g_n = J_n^\top r_n,$$

where $\bar{h}_n \geq 0$ has empirical mean one. The continual-learning penalty is

$$\frac{\lambda}{2} (\theta - \theta_A^\star)^\top M_{\mathrm{IEWC}} (\theta - \theta_A^\star).$$

The canonical geometry is $G_n = I$. We use task-relative damping

$$\tau_A = \kappa \ \mathrm{median}\{r_i^\top G_i^{-1} r_i : r_i \neq 0\},$$

with $\kappa = 10^{-2}$ in the confirmatory comparison. This makes κ dimensionless and preserves global loss-unit invariance. If all denominators are zero, the stored importance is zero.

Uniform IEWC sets $\bar{h}_n = 1$. To study weighting without confounding it with normalization, we define a separate *residual-diversity weighting* (RDW) variant. Let

$$u_n = \frac{g_n}{\|g_n\|_2}$$

for nonzero parameter gradients. In a fixed sample order, RDW maintains an orthonormal basis Q of at most K earlier residual directions and assigns

$$h_n = \|(I - Q^\top Q) u_n\|_2^2.$$

When the residual is nonzero and the cap is not reached, its normalized direction is appended to Q; the final h_n are mean-normalized. This streaming rule is motivated by residual-gradient diversity ideas in GSS [1], but it is an order-dependent aggregation heuristic, not part of the level-set derivation. Crucially, both OPG-RDW and IEF-RDW receive the same unit parameter-gradient directions. Scalar IEF normalization cannot change u_n, so the 2×2 comparison

$$\{\mathrm{OPG}, \mathrm{IEF}\} \times \{\mathrm{uniform}, \mathrm{RDW}\}$$

isolates matrix normalization from sample weighting.

For diagonal training penalties, we take the diagonal after aggregation and rescale every nonzero diagonal to have mean one over trainable parameters. This removes only a global factor absorbable into λ and lets all regularized methods use the same within-domain λ grid. It does not restore the cross-parameter information discarded by diagonalization.

The design can be summarized by $(\lambda, \kappa, \Gamma, h)$, where Γ maps an old output to its local metric G_n, and by an additional approximation choice such as diagonal or low rank. Scalar loss preservation gives OPG-EWC; level-set normal preservation gives uniform IEWC; damping, weighting, and geometry are explicit surrounding choices. We call this CLOC-derived mechanism *Improved Elastic Weight Consolidation* while treating its individual axes separately in theory and experiment.

15 Empirical Evaluation

We use two evidence tiers. The confirmatory comparison is frozen before locked-test access and asks whether IEF normalization helps relative to OPG under identical uniform weights and matched global diagonal scale. It uses disjoint train, validation, and locked-test roles; paired task orders and optimizer seeds; fixed five-point validation grids without edge expansion; and paired uncertainty on the held-out repetitions. The secondary 2×2 comparison crosses OPG/IEF with uniform/RDW aggregation. Sampled model-Fisher EWC and MAS are external regularization baselines; sequential training and offline/joint training provide lower-context and non-continual references. This comparator set preserves the same post-task diagonal-penalty interface: SI's trajectory accumulation and LwF's distillation objective are discussed in Section 2 but are not mixed into the estimator-focused factorial. Appendix C specifies the complete protocol.

The original multi-domain study is retained quantitatively in full because it establishes IEWC's empirical breadth and attained performance across classification, NLP, forecasting, segmentation, and diffusion. Its selected endpoints participated in hyperparameter and method development, so comparisons within that study are descriptive rather than selection-independent effect estimates; the new locked study supplies the latter evidence under a matched factorial design. Unless stated otherwise, all stored matrices are diagonal. Historical labels embedded in the original figures map as follows: EF-EWC denotes observed-label OPG-EWC, IEWC-E or U-IEWC denotes uniform IEF, and IEWC-GSS or GSS-IEWC denotes the earlier GSS-motivated weighting implementation. The last implementation is order dependent, so its gains cannot be attributed to normalization alone.

15.1 Established multi-domain results

Task type	Metric	Sequential	OPG-EWC	EWC-DR	U-IEWC	GSS-IEWC
Classification	Final avg. accuracy ↑	0.2896 ± 0.0405	0.5981 ± 0.0337	0.5687 ± 0.0263	**0.6332 ± 0.0232**	0.6250 ± 0.0263
Classification	Avg. forgetting ↓	0.7066 ± 0.0466	0.3595 ± 0.0378	0.3933 ± 0.0286	**0.3142 ± 0.0250**	0.3285 ± 0.0278
NLP	Final avg. score ↑	0.5760 ± 0.0027	0.5972 ± 0.0020	0.5972 ± 0.0059	0.5938 ± 0.0034	**0.6272 ± 0.0080**
NLP	Avg. forgetting ↓	0.1139 ± 0.0111	0.0762 ± 0.0034	0.0781 ± 0.0109	0.0820 ± 0.0122	**0.0046 ± 0.0079**
Forecasting	Final avg. MSE ↓	0.2733 ± 0.0065	0.2388 ± 0.0021	–	0.2418 ± 0.0038	**0.2383 ± 0.0007**
Forecasting	Avg. MSE forgetting ↓	0.0535 ± 0.0081	**0.0140 ± 0.0020**	–	0.0191 ± 0.0047	0.0147 ± 0.0021
Segmentation	Final all mIoU ↑	0.0917 ± 0.0061	0.0946 ± 0.0040	0.3080 ± 0.0307	**0.4678 ± 0.0568**	0.4274 ± 0.0574
Segmentation	Base-class forgetting ↓	0.7135 ± 0.0114	0.7131 ± 0.0166	0.4471 ± 0.0447	**0.2418 ± 0.0965**	0.2925 ± 0.0839
Diffusion	Final avg. denoising MSE ↓	0.0676 ± 0.0029	0.0296 ± 0.0013	–	0.0298 ± 0.0014	**0.0293 ± 0.0013**
Diffusion	Denoising-MSE forgetting ↓	0.0503 ± 0.0002	**-0.0006 ± 0.0020**	–	0.0001 ± 0.0024	-0.0003 ± 0.0020

Table 2: Results from the original multi-domain study, reported as selected-run means and across-seed standard deviations. The table establishes attained performance across five task types; because the endpoints participated in development, pairwise differences are descriptive rather than selection-independent estimates. Higher is better for accuracy, score, and mIoU; lower is better for MSE and forgetting.

Table 2 shows that uniform IEWC attained higher selected means than OPG-EWC on ImageNet-R and VOC 15–5, while the weighted IEWC configuration attained the highest TRACE score and the lowest forecasting and DDPM MSE. Together, these results demonstrate broad applicability and strong selected performance. The matched factorial and locked evaluation then isolates which gains generalize and which arise from normalization versus weighting.

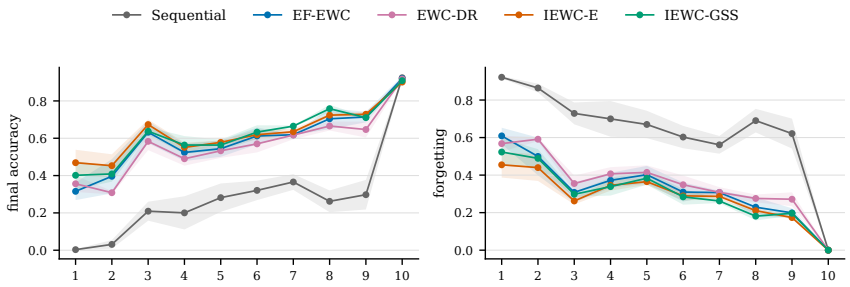


Figure 2: ImageNet-R retention in the original multi-domain study. The left panel shows final accuracy per distribution; the right shows forgetting.

At the distribution level, Figure 2 shows that IEWC's retained accuracy is not concentrated in a single late distribution. The locked paired analysis tests whether this advantage generalizes across newly frozen class orders.

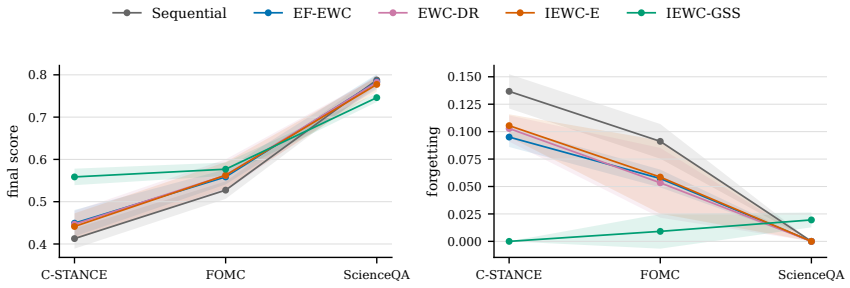


Figure 3: TRACE retention for C-STANCE, FOMC, and ScienceQA in the original study. FOMC's distributed evaluation and test files are duplicates, so the new locked comparison uses C-STANCE, ScienceQA, and NumGLUE-cm.

The weighted IEWC run likewise has less forgetting on C-STANCE and FOMC in Figure 3. The revised comparison retains the weighting mechanism but crosses it with both OPG and IEF, while replacing FOMC with NumGLUE-cm to obtain disjoint validation and test roles.

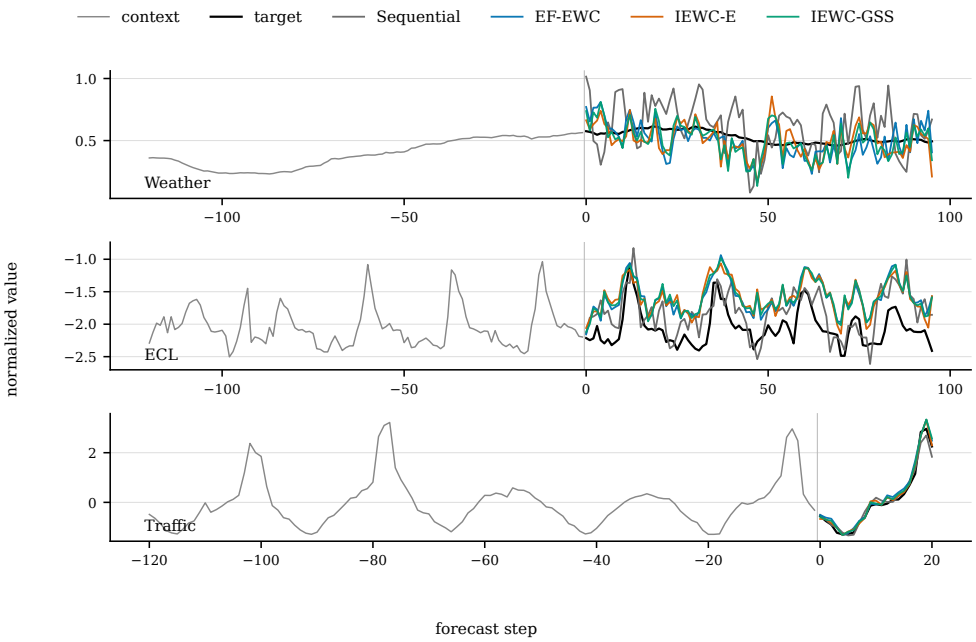


Figure 4: Forecasting rollout comparison. Each panel shows one held-out rollout from the corresponding forecasting distribution. The gray curve is the observed context, the black curve is the target horizon, and colored curves are method predictions after the full continual sequence.

Figure 4 complements the aggregate forecasting scores with individual held-out rollouts. Weighted IEWC retains more target-trajectory structure at the first task and remains close to OPG-EWC later; these examples illustrate the observed behavior rather than replace the aggregate uncertainty estimates.

The DDPM study also tested an alternative output representation originally labeled "sliced Wasserstein." The selected run reduced old-output drift to 9.4409 ± 7.2517, compared with 14.1026 ± 12.2220 for Euclidean uniform IEWC, while attaining denoising MSE 0.0301 ± 0.0012 versus 0.0298 ± 0.0014. Because that surrogate did not establish a matched primal/dual metric pair, the result supports output-representation flexibility but not a claim about Wasserstein geometry. Section 11 supplies a correct ordered-CDF/Cramér construction.

The FROMP-derived weighting ablation [16] reached 0.6365 ± 0.0165 final average accuracy and 0.3068 ± 0.0170 forgetting on ImageNet-R, but 0.3351 ± 0.0742 final all-mIoU and 0.4064 ± 0.1122 base-class forgetting on VOC 15–5. Its domain-dependent behavior further motivates treating aggregation as an explicit design axis rather than folding it into the normalization claim.

Appendix C records both the original workflow and the frozen confirmatory protocol, together with damping and effective-summand diagnostics. Appendix D reports the downstream label-noise stress test, while Appendix B supplies the new direct clean/corrupt contribution analysis.

Code Release

We release an accompanying repository with a readily usable IEWC implementation. The `IEWCConfig` class encapsulates the four-tuple parametrization introduced above. The same configuration interface is used by the diagonal and low-rank IEWC paths and by the empirical scripts, including the FACIL-based CIFAR-100 checks and the larger ImageNet-R, TRACE, forecasting, segmentation, and diffusion runs. The repository is available at https://github.com/Axym-Labs/iewc.

16 Limitations

CLOC is a formulation of preservation, not a claim that all old-task behavior can be retained. Per-sample equality is stronger than expected-risk preservation and can conflict with fitting later tasks. The penalty coefficient therefore controls a stability–plasticity relaxation rather than guaranteeing feasibility of the original hard constraint.

The OPG and IEF identities are local. They follow after linearizing the chosen scalar or output map at the old solution and squaring the resulting first-order change. Neural-network loss level sets are generally nonconvex, their global projections need not be unique, and the tangent approximation can become inaccurate after a large parameter displacement. At a stationary output loss, the level-set normal is undefined; finite damping assigns zero contribution and restores attenuation in the near-zero regime.

The output metric must be specified as a matched primal/dual pair. Ambient singular metrics require restriction to an identifiable subspace and a stated pseudoinverse convention. Our ordered-CDF construction satisfies that requirement on the probability-simplex tangent, but the inherited diffusion result is exploratory and does not validate a general Wasserstein geometry.

Practical IEWC also introduces approximations beyond the derivation. A diagonal retains coordinatewise importance but discards correlations, eigenspaces, and reparameterization behavior. Importance estimates use finitely many old samples. RDW is an order-dependent weighting heuristic with additional cost and is not implied by level-set normalization. Relative damping guarantees invariance to a global change of loss units, not to arbitrary sample-dependent transformations at finite damping.

Finally, empirical comparisons cover selected datasets, model families, adaptation regimes, and task orders. Validation selection and locked paired evaluation reduce selection bias but cannot establish a domain-independent advantage or deployment robustness. Offline/joint training is only a reference because it violates the continual data-access constraint, and untested replay, distillation, and parameter-isolation methods may be preferable when their storage or architectural assumptions are allowed.

17 Conclusion

We formulated continual learning as an old-solution preservation constraint (CLOC) and derived two local quadratic relaxations. Preserving each old sample's scalar loss yields an outer-product loss-gradient matrix; for negative log likelihood, this is algebraically the observed-label empirical Fisher used in many EWC implementations, but it is not the model Fisher or a general curvature identity. Preserving outputs modulo the tangent to the old loss level set instead yields the iEF normalization. This retains the central geometric intuition while locating its exact boundary: undamped normalization removes the samplewise output-loss-gradient scale, whereas positive damping multiplies each summand by $s/(s+\tau)$ and assigns zero contribution at exact interpolation.

Task-relative damping restores invariance to global changes of loss units, and the controlled contamination study directly shows that normalization can prevent high-loss corrupted samples from dominating a full importance matrix in the tested construction. Whether this altered local geometry improves a diagonal continual-learning penalty remains an empirical, domain-conditional question. Our evaluation therefore separates normalization from residual-diversity weighting, matches global scale, includes external regularization and offline references, selects hyperparameters only on validation data, and reserves paired task orders and test splits for locked inference. IEWC is therefore a preservation-first CLOC family: its explicit damping, weighting, output metric, and matrix approximation explain normalized importance geometrically and guide new continual-learning penalties.

References

[1] Rahaf Aljundi, Min Lin, Baptiste Goujaud, and Yoshua Bengio. Gradient based sample selection for online continual learning. In *NeurIPS*, 2019.

[2] Michael McCloskey and Neal J. Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of Learning and Motivation*, volume 24. Academic Press, 1989.

[3] Robert M. French. Catastrophic forgetting in connectionist networks. *Trends in Cognitive Sciences*, 3(4), 1999.

[4] German I. Parisi, Ronald Kemker, Jose L. Part, Christopher Kanan, and Stefan Wermter. Continual lifelong learning with neural networks: A review. *Neural Networks*, 113, 2019.

[5] Gido M. van de Ven and Andreas S. Tolias. Three scenarios for continual learning, 2019.

[6] James Kirkpatrick et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 2017.

[7] Friedemann Zenke, Ben Poole, and Surya Ganguli. Continual learning through synaptic intelligence. In *ICML*, 2017.

[8] Rahaf Aljundi, Francesca Babiloni, Mohamed Elhoseiny, Marcus Rohrbach, and Tinne Tuytelaars. Memory aware synapses: Learning what (not) to forget. In *ECCV*, 2018.

[9] Zhizhong Li and Derek Hoiem. Learning without forgetting. In *ECCV*, 2016.

[10] Ferenc Huszar. Note on the quadratic penalties in elastic weight consolidation. *Proceedings of the National Academy of Sciences*, 115(11), 2018.

[11] Jonathan Schwarz, Jelena Luketina, Wojciech M. Czarnecki, Agnieszka Grabska-Barwinska, Yee Whye Teh, Razvan Pascanu, and Raia Hadsell. Progress & compress: A scalable framework for continual learning. In *ICML*, 2018.

[12] Cuong V. Nguyen, Yingzhen Li, Thang D. Bui, and Richard E. Turner. Variational continual learning. In *ICLR*, 2018.

[13] Frederik Kunstner, Lukas Balles, and Philipp Hennig. Limitations of the empirical fisher approximation for natural gradient descent. In *NeurIPS*, 2019.

[14] Gido M. van de Ven. On the computation of the fisher information in continual learning, 2025.

[15] Michalis K. Titsias, Jonathan Schwarz, Alexander G. de G. Matthews, Razvan Pascanu, and Yee Whye Teh. Functional regularisation for continual learning with gaussian processes. In *ICLR*, 2020.

[16] Pingbo Pan, Siddharth Swaroop, Alexander Immer, Runa Eschenhagen, Richard E. Turner, and Mohammad Emtiyaz Khan. Continual deep learning by functional regularisation of memorable past. In *NeurIPS*, 2020.

[17] X. Wu, W. Yu, C. Zhang, and P. Woodland. An improved empirical fisher approximation for natural gradient descent. In *Advances in Neural Information Processing Systems*, 2024.

[18] Neal Parikh and Stephen Boyd. Proximal algorithms. *Foundations and Trends in Optimization*, 1 (3), 2014.

[19] David Lopez-Paz and Marc'Aurelio Ranzato. Gradient episodic memory for continual learning. In *NeurIPS*, 2017.

[20] Arslan Chaudhry, Marc'Aurelio Ranzato, Marcus Rohrbach, and Mohamed Elhoseiny. Efficient lifelong learning with a-gem. In *ICLR*, 2019.

[21] Shun-ichi Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10(2), 1998.

[22] James Martens. New insights and perspectives on the natural gradient method. *Journal of Machine Learning Research*, 21, 2020.

[23] James Martens and Roger Grosse. Optimizing neural networks with kronecker-factored approximate curvature. In *ICML*, 2015.

[24] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network, 2015.

[25] Nicol N. Schraudolph. Fast curvature matrix-vector products for second-order gradient descent. *Neural Computation*, 14(7), 2002.

[26] James Martens. Deep learning via hessian-free optimization. In *ICML*, 2010.

[27] X. Liu and X. Chang. Elastic weight consolidation done right for continual learning. In *CVPR*, 2026.

[28] Mikhail Belkin, Daniel Hsu, Siyuan Ma, and Soumik Mandal. Reconciling modern machine learning practice and the bias–variance trade-off. *Proceedings of the National Academy of Sciences*, 2019.

A Invariance to Loss Reparameterization

This appendix treats the undamped case $\tau = 0$.

The output-level feasible set

$$\mathcal{C}_n = \{z : \ell_n(z) = \ell_n(z_n^A)\}$$

depends on the level sets of ℓ_n, not on the numerical labels assigned to those level sets. If we apply a local monotone scalar transformation

$$\tilde{\ell} = \phi(\ell), \qquad \phi'(\ell) > 0,$$

then the output gradient rescales as

$$\tilde{r}_n = \phi'(\ell_n(z_n^A))\, r_n.$$

For OPG, the sample contribution changes by

$$J_n^\top \tilde{r}_n \tilde{r}_n^\top J_n = \phi'(\ell_n(z_n^A))^2 J_n^\top r_n r_n^\top J_n.$$

For IEF, the same scalar factor appears in numerator and denominator:

$$\frac{\tilde{r}_n \tilde{r}_n^\top}{\tilde{r}_n^\top G_n^{-1} \tilde{r}_n} = \frac{r_n r_n^\top}{r_n^\top G_n^{-1} r_n}.$$

Thus undamped IEF is invariant to local scalar relabelings of the constraint function wherever $r_n \neq 0$. This includes loss rescaling, unit changes in regression losses, reduction conventions, and smooth monotone transformations that preserve the local level-set geometry. OPG's global scale can be absorbed into λ only when the transformation rescales all samples equally; sample-dependent factors change its relative weighting.

Positive shared damping narrows the invariance. A fixed numerical τ does not transform with the loss and therefore changes even under a global unit rescaling. Task-relative damping $\tau_A = \kappa \operatorname{median}_{i:s_i>0} s_i$ restores invariance to a global task-loss multiplier because s_i and τ_A scale together. It does not generally preserve invariance to sample-dependent monotone transformations. At $r_n = 0$, the implementation assigns a zero contribution rather than claiming an undamped $0/0$ identity.

B Contamination Contributions and Spectral Stability

The OPG–IEF normalization difference can change which samples dominate a full importance matrix. In Euclidean output geometry, write $s_i = \|r_i\|_2^2$ and $a_i = g_i/\sqrt{s_i}$ for $s_i > 0$. The undamped per-sample matrices are

$$M_{\mathrm{OPG},i} = s_i a_i a_i^\top, \qquad M_{\mathrm{IEF},i} = a_i a_i^\top.$$

Thus OPG couples a sensitivity direction to the squared output-loss-gradient scale, whereas undamped IEF keeps the direction-only outer product. Relative damping multiplies the latter by $s_i/(s_i + \tau_A)$ and therefore retains the near-zero qualification from Section 10.

For intuition, suppose a fraction ε of samples are contaminated, their normalized sensitivity is concentrated near a direction u outside the leading clean subspace, and their scale is $s_i \approx W$. If clean scales are sufficiently concentrated or independent of direction, then

$$\mathbb{E}[M_{\mathrm{OPG}}] \approx (1-\varepsilon)\bar{s}_{\mathrm{clean}} \Sigma_{\mathrm{clean}} + \varepsilon W u u^\top,$$

whereas undamped IEF has no factor W on the contaminating direction. In particular, a sufficiently large εW can move u into the leading OPG eigenspace. This model is illustrative rather than a distribution-free robustness theorem.

We test the mechanism directly in a fitted two-class MLP. For each of 20 independent seeds and label-flip rates 0.05, 0.10, and 0.20, we retain all 512 per-sample parameter-gradient rows and clean/-corrupt identities. We decompose the full OPG and relative-damped IEF matrices into clean and corrupt contributions, then compare each total matrix with its clean-contribution reference. Table 3 reports means over seeds.

Flip rate	Corrupt/clean s_i ratio	Corrupt trace share		Leading-vector alignment		Top-5 overlap	
		OPG	IEF	OPG	IEF	OPG	IEF
0.05	81.26	0.783	0.054	0.426	1.000	0.902	0.996
0.10	36.67	0.789	0.104	0.720	0.999	0.923	0.997
0.20	11.23	0.736	0.202	0.875	0.998	0.930	0.990

Table 3: Direct full-matrix contamination diagnostic. Alignment is the absolute inner product of the leading eigenvectors; top-5 overlap is the mean squared singular value between the corresponding eigenspaces. Each reference is the clean contribution from the same fitted model.

Paired 95% intervals over seeds exclude zero for every IEF-minus-OPG contrast in the table. Across increasing flip rates, the trace-share differences are -0.729, -0.685, and -0.534; leading-vector alignment differences are $+0.574$, $+0.279$, and $+0.123$; and top-5-overlap differences are $+0.094$, $+0.074$, and $+0.060$. These observations support the proposed contamination mechanism in this controlled construction. They do not establish a universal robustness guarantee or prove that a diagonal continual-learning penalty improves on a real benchmark; those are separate empirical questions.

C Experimental Protocol, Provenance, and Sensitivity

C.1 Frozen confirmatory protocol

The confirmatory protocol was serialized and hashed before any new locked-test model outcome was computed. It fixes data roles, methods, grids, repetitions, estimands, uncertainty calculations, and stopping rules. Locked-test loading requires the manifest hash and writes an append-only access record before loading examples. Validation scripts reject a locked-test role, and validation repetitions are excluded from confirmatory intervals.

The primary contrast in each domain is paired final average performance of uniform IEF minus uniform OPG. The secondary factorial comparison is

$$\{\text{OPG}, \text{IEF}\} \times \{\text{uniform}, \text{RDW}\}.$$

All four cells use the same retained examples, and the RDW cells receive the same unit-normalized parameter-gradient directions. Sampled model-Fisher EWC and MAS [8] are external regularization baselines. Sequential and offline/joint training are references. Every nonzero diagonal importance is rescaled to mean one across trainable parameters, and every method uses the displayed $\lambda/2$ convention.

Domain	Data roles	Validation runs	Locked runs
ImageNet-R	Stratified 80%/20% split of the original 24,000-example train set for train/locked test; the previously accessed 6,000-example test set is validation only	3 class orders	10 paired class orders
TRACE	Distributed train/eval/test files for C-STANCE, ScienceQA, and NumGLUE-cm; prompt/answer overlaps are zero	3 task orders	6 task orders $\times$ 2 optimizer seeds

Table 4: Frozen confirmatory roles and repetitions. FOMC is excluded because its distributed evaluation and test files are exact duplicates.

ImageNet-R uses the fixed λ grid $\{0.01, 0.03, 0.1, 0.3, 1\}$. TRACE uses $10^{-5} \times \{1, 3, 10, 30, 100\}$. Neither grid permits edge expansion.

ImageNet-R uses a pretrained ViT-B/16 with LoRA rank 4 and ten 20-class tasks, trained for two epochs per task with AdamW at learning rate $3 \cdot 10^{-4}$ and batch size 64. Importance uses 128 examples per completed task and RDW cap 64. TRACE uses Qwen2.5-3B-Instruct with LoRA rank 8, two epochs, AdamW at $2 \cdot 10^{-4}$, batch size 2 with gradient accumulation 8, 512 training examples per task, and 64 importance examples per completed task. Data, loader, optimizer, task-order, importance-order, and sampled-Fisher-label seeds are separately recorded and paired where appropriate.

Confirmatory execution uses explicit DataLoader generators, reseeds each task's training stage independently of intervening importance computation, and requires deterministic PyTorch and CUDA kernels. Before restarting validation, two complete same-configuration runs in each domain were

compared after removing wall-clock metadata. Their training losses, evaluation matrices, and, for TRACE, generated example-level predictions matched exactly. These duplicate checks are retained for reproducibility auditing but excluded from selection.

For every paired contrast we report the mean and standard deviation of paired differences, a two-sided 95% Student-t interval, the exact two-sided sign-flip randomization p-value when enumerable, the median, and every paired point. These summaries treat task-order/seed pairs as independent; the interval assumes approximately normal paired differences, and sign-flip validity assumes sign exchangeability under the null. The primary interpretation uses a predeclared practical-difference threshold of 0.01 rather than a binary significance rule. A pre-lock audit found that the frozen protocol listed five secondary contrasts but called them four; the dated clarification applies Holm correction to all five within a domain. Full five-point validation curves, importance time, total run time, and peak CUDA memory are retained.

IEF uses task-relative damping

$$\tau_A = 10^{-2}\,\mathrm{median}\{s_i : s_i > 0\}.$$

Validation-only sensitivity varies the coefficient over $\{0, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$, importance counts, RDW caps, and RDW sample order. A separate loss-unit diagnostic multiplies a fitted classification loss by 0.1, 1, and 10. OPG raw scale changes quadratically, fixed numerical damping changes IEF shape, and task-relative damping preserves the IEF matrix to numerical tolerance, matching Section 10.

C.2 Original multi-domain protocol

Table 5 records the configurations behind Table 2. These values are retained for exact provenance. In the original study, final reported endpoints participated in grid centering, edge expansion, and method selection; the runs therefore do not have an untouched selection-independent endpoint. The original implementation also omitted the displayed factor $1/2$, so its numerical λ values should not be compared directly with the revised convention.

Task type	Dataset, sequence, and model	Originally selected values
Classification	ImageNet-R; ten 20-class distributions; pretrained ViT-B/16 with LoRA rank 4.	OPG: $\lambda = 333.333$; DR: $\lambda = 3$; U-IEWC: $\lambda = 33.333, \tau = 10^{-4}$; weighted IEWC: $\lambda = 11.111, \tau = 10^{-2}$.
NLP	TRACE, C-STANCE $\rightarrow$ FOMC $\rightarrow$ ScienceQA; pretrained Qwen2.5-3B with LoRA rank 8.	OPG: $\lambda = 1$; DR: $\lambda = 1$; U-IEWC: $\lambda = 0.111, \tau = 10^{-2}$; weighted IEWC: $\lambda = 59049, \tau = 10^{-2}$.
Forecasting	Weather $\rightarrow$ ECL $\rightarrow$ Traffic; PatchTST trained from scratch.	OPG: $\lambda = 729$; U-IEWC: $\lambda = 9, \tau = 10^{-2}$; weighted IEWC: $\lambda = 9, \tau = 10^{-2}$.
Segmentation	VOC 15–5, base then novel classes; pretrained ConvNeXtV2-tiny FPN with full fine-tuning.	OPG: $\lambda = 0.111$; DR: $\lambda = 243$; U-IEWC: $\lambda = 270, \tau = 10^{-3}$; weighted IEWC: $\lambda = 2430, \tau = 10^{-2}$.
Diffusion	MNIST DDPM, labels $0/1 \rightarrow 2/3$; conditional DDPM trained from scratch.	OPG: $\lambda = 9 \cdot 10^5$; U-IEWC: $\lambda = 675, \tau = 10^{-3}$; weighted IEWC: $\lambda = 225, \tau = 10^{-3}$; output ablation: $\lambda = 10^6, \tau = 10^{-3}$.

Table 5: Configurations for the original multi-domain study. "DR" denotes EWC-DR. These hyperparameters belong to the original implementation and selection workflow.

The original search used a five-point multiplicative grid around a task-specific midpoint and extended an edge up to three times when an endpoint was best. This workflow supports descriptive selected-performance claims but not a selection-independent effect estimate. The revised protocol

fixes all five points before validation, forbids edge expansion, and accesses the locked test only after a selection artifact is frozen.

C.3 Damping and summand diagnostics

Figure 5 reports the fixed-τ sensitivity experiment on a three-task CIFAR-100/FACIL prefix. It demonstrates that damping is consequential, while also showing why a universal numerical default is inappropriate: τ has loss-dependent units. The revised experiment therefore uses the dimensionless relative coefficient κ.

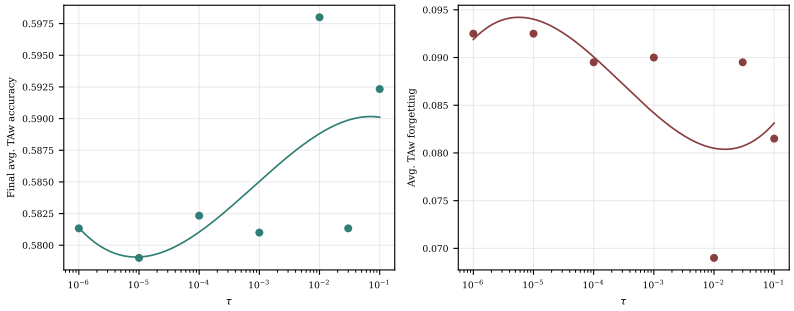


Figure 5: Fixed-τ sensitivity on a CIFAR-100 prefix. The tested numerical values were $\{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}, 10^{-1}\}$. The result demonstrates the practical importance of reporting damping explicitly.

Figure 6 reports the diagonal scale diagnostic for CIFAR-100 classification, controlled regression, MNIST DDPM denoising, and binary segmentation. It shows λ times the per-sample diagonal trace before averaging and is not a sample-weight plot.

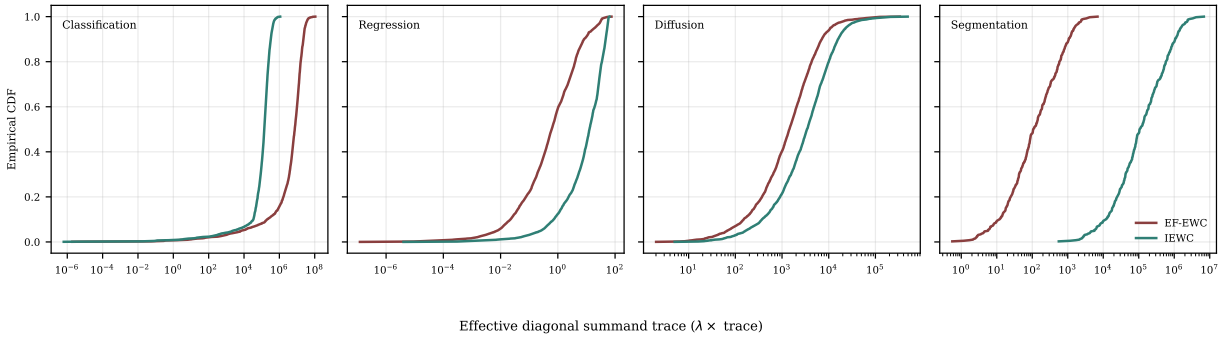


Figure 6: Effective diagonal summand norms. Each panel reports the empirical CDF of λ times the per-sample diagonal trace in the original configuration.

The earlier EWC-DR check additionally tested $\lambda \in \{100, 300, 1000, 3000, 10000\}$ on that CIFAR-100 prefix and selected 100. It remains a local historical sensitivity check, not part of the ImageNet-R confirmatory selection.

D Label-Noise Stress Test

The original CIFAR-100 label-noise experiment provides a downstream stress test complementary to the direct matrix analysis in Appendix B. Its endpoints were available during method development,

so the table reports descriptive selected performance rather than a selection-independent effect estimate. Uniform training-label flips create atypical high-loss samples, and the clean task-aware test set measures downstream retention. Across this stress test, IEWC loses less accuracy than observed-label OPG-EWC as the flip rate increases.

Method	$p = 0$	$p = 0.1$	$p = 0.25$
OPG-EWC	0.5617 ± 0.0062	0.5132 ± 0.0112	0.2773 ± 0.0320
EWC-DR	0.5932 ± 0.0296	0.5310 ± 0.0226	0.4598 ± 0.0095
IEWC	0.5978 ± 0.0260	0.5642 ± 0.0215	0.5322 ± 0.0239

The original study also compared sorted diagonal entries on a CIFAR-100 prefix. That diagnostic does not identify clean and corrupt sample contributions and cannot establish the spectral mechanism by itself; Appendix B now supplies the direct per-sample full-matrix test. The diagonal summary remains useful as a complementary downstream view. It uses all 512 exported entries per matrix and reports the stored-tail mass ratio, normalized stored-tail profile change, and log-scale RMSE of the sorted entries.

Method	Noise	Entries	Tail mass ratio	Profile L1	Log-scale RMSE
OPG-EWC	0.1	512	0.0288	0.2610	1.4660
OPG-EWC	0.25	512	0.0244	0.2737	1.5337
IEWC	0.1	512	1.0833	0.0350	0.0350
IEWC	0.25	512	1.1003	0.0154	0.0415

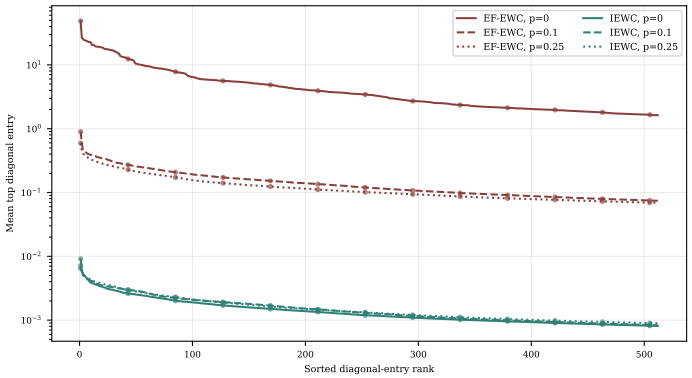


Figure 7: Label-noise diagonal diagnostic. The plot visualizes the sorted high-importance coordinate tail and complements the direct clean/corrupt contribution analysis in Appendix B.